

L12



L12: The Legendre Symbol

①

Definition: Let p be an odd prime number and let $a \in \mathbb{Z}$ with $p \nmid a$. The Legendre symbol, denoted $\left(\frac{a}{p}\right)$, is

$$\left(\frac{a}{p}\right) := \begin{cases} 1 & \text{if } a \text{ is a quadratic residue mod } p \\ -1 & \text{if } a \text{ is a quadratic non residue mod } p \end{cases}.$$

Example: $1, 3, 4, 5, 9$ are quadratic residues modulo 11. Thus

$$\left(\frac{1}{11}\right) = \left(\frac{3}{11}\right) = \left(\frac{4}{11}\right) = \left(\frac{5}{11}\right) = \left(\frac{9}{11}\right) = 1.$$

$2, 6, 7, 8, 10$ are quadratic non-residues modulo 11. Thus

$$\left(\frac{2}{11}\right) = \left(\frac{6}{11}\right) = \left(\frac{7}{11}\right) = \left(\frac{8}{11}\right) = \left(\frac{10}{11}\right) = -1.$$

• The Legendre symbol $\left(\frac{a}{p}\right)$ indicates whether a is a quadratic residue modulo p or not (in other words, whether $x^2 \equiv a \pmod{p}$ is Solvable or not).

- The Legendre symbol does not tell you the actual solutions to the congruence, these need to be found by other means. (2)

Example: Find the value of $(\frac{3}{7})$. The Legendre symbol $(\frac{3}{7})$ asks the question: is the congruence $x^2 \equiv 3 \pmod{7}$ solvable? We have

$$1^2 \equiv 1 \pmod{7} \Rightarrow 1 \text{ is a quadratic residue } \pmod{7}$$

$$2^2 \equiv 4 \pmod{7} \Rightarrow 4 \text{ is a quadratic residue } \pmod{7}$$

$$3^2 \equiv 2 \pmod{7} \Rightarrow 2 \text{ is a quadratic residue } \pmod{7}$$

By Proposition 4.3, there are exactly

$$\frac{p-1}{2} = \frac{7-1}{2} = 3 \text{ incongruent quadratic residues}$$

modulo 7. Since 1, 2, 4 are quadratic residues

modulo 7, then 3, 5, 6 must be quadratic

non-residues modulo 7. Thus $x^2 \equiv 3 \pmod{7}$

is not solvable, and so $(\frac{3}{7}) = -1$.

- The only method covered so far to compute $(\frac{a}{p})$ is the computation of $\frac{p-1}{2}$ congruences. We will improve our tool kit!

(3)

Theorem 4.4 (Euler's Criterion): Let p be an odd prime number and let $a \in \mathbb{Z}$ with $p \nmid a$. Then

$$\left(\frac{a}{p}\right) \equiv a^{(p-1)/2} \pmod{p}.$$

Proof: Assume that $\left(\frac{a}{p}\right) = 1$. Then $x^2 \equiv a \pmod{p}$ has a solution, say x_0 . We have

$$a^{(p-1)/2} \equiv \left(x_0^2\right)^{(p-1)/2} \pmod{p}$$

$$\equiv x_0^{p-1} \pmod{p}$$

$$\equiv 1 \pmod{p} \text{ by Fermat's Little Theorem (since } p \nmid x_0)$$

$$\equiv \left(\frac{a}{p}\right) \pmod{p}.$$

$$\equiv 1 \pmod{p}.$$

Thus the desired result holds if $\left(\frac{a}{p}\right) = 1$.

Now assume that $\left(\frac{a}{p}\right) = -1$. For each $i \in \mathbb{Z}$ with $1 \leq i \leq p-1$, there is a unique $j \in \mathbb{Z}$ with $1 \leq j \leq p-1$ such that $ij \equiv a \pmod{p}$

by Theorem 2.6. Furthermore, $i \neq j$, since (4) if $i=j$, then $i^2 \equiv a \pmod{p}$ implies that a is a quadratic residue modulo p and we obtain $\left(\frac{a}{p}\right) = 1$, a contradiction! Thus we can pair the integers into $1, 2, \dots, p-1$ into $\frac{p-1}{2}$ pairs, each with product a modulo p . Then

$$(p-1)! \equiv a^{(p-1)/2} \pmod{p}.$$

Thus Wilson's Theorem implies that

$$a^{(p-1)/2} \equiv -1 \pmod{p},$$

from which we obtain

$$\left(\frac{a}{p}\right) \equiv a^{(p-1)/2} \pmod{p},$$

and the desired result holds if $\left(\frac{a}{p}\right) = -1$.

Example: Find $\left(\frac{3}{7}\right)$ using Euler's criterion.

$$\text{We have } \left(\frac{3}{7}\right) = 3^{(7-1)/2} \pmod{7}$$

$$\equiv 27 \pmod{7}$$

$$\equiv -1 \pmod{7}.$$

$$\text{Thus } \left(\frac{3}{7}\right) = -1.$$

(5)

Proposition 4.5: Let p be an odd prime, and let $a, b \in \mathbb{Z}$ with $p \nmid a$ and $p \nmid b$. Then

$$(a) \left(\frac{a^2}{p} \right) = 1$$

$$(b) \text{ If } a \equiv b \pmod{p}, \text{ then } \left(\frac{a}{p} \right) = \left(\frac{b}{p} \right)$$

$$(c) \left(\frac{ab}{p} \right) = \left(\frac{a}{p} \right) \left(\frac{b}{p} \right).$$

Proof: (a) The congruence $x^2 \equiv a^2 \pmod{p}$ has a solution, namely a .

(b) Since $a \equiv b \pmod{p}$, the quadratic congruence to be solved is $x^2 \equiv a \equiv b \pmod{p}$. Clearly $x^2 \equiv a \pmod{p}$ is solvable if and only if $x^2 \equiv b \pmod{p}$ is solvable.

$$\begin{aligned} (c) \text{ We have } \left(\frac{ab}{p} \right) &\equiv (ab)^{(p-1)/2} \pmod{p} \text{ by Euler's criterion} \\ &\equiv a^{(p-1)/2} b^{(p-1)/2} \pmod{p} \\ &\equiv \left(\frac{a}{p} \right) \left(\frac{b}{p} \right) \pmod{p}. \end{aligned}$$

Since the only values of the Legendre Symbol

are ± 1 , we have

(6)

$$\left(\frac{ab}{p}\right) = \left(\frac{a}{p}\right)\left(\frac{b}{p}\right).$$

(3)

Example: Find $\left(\frac{-11}{7}\right)$. We have

$$\left(\frac{-11}{7}\right) = \left(\frac{-1}{7}\right)\left(\frac{11}{7}\right) = \left(\frac{-1}{7}\right)\left(\frac{4}{7}\right) = \left(\frac{-1}{7}\right)$$

$$= \left(\frac{6}{7}\right) = -1, \text{ where the final equality}$$

follows from the previous example.

• Further properties of Legendre symbols: let p be an odd prime and $a \in \mathbb{Z}$ with $p \nmid a$.

Then $a = \pm 2^{a_0} p_1^{a_1} p_2^{a_2} \cdots p_r^{a_r}$ with $p_1, p_2, \dots, p_r$

odd primes distinct from p and $a_1, a_2, \dots, a_r$

non-negative integers. By Proposition 4.5(c)

we have that

$$\left(\frac{a}{p}\right) = \left(\frac{\pm 1}{p}\right) \left(\frac{2}{p}\right)^{a_0} \left(\frac{p_1}{p}\right)^{a_1} \left(\frac{p_2}{p}\right)^{a_2} \cdots \left(\frac{p_r}{p}\right)^{a_r}.$$

Thus it suffices to compute $\left(\frac{\pm 1}{p}\right)$, $\left(\frac{2}{p}\right)$, and $\left(\frac{q}{p}\right)$

for q an odd prime distinct from p .

(7)

Plan: • $\left(\frac{1}{p}\right) = 1$ by Proposition 4.5(a).

- Next will compute $\left(\frac{-1}{p}\right)$
- Consider $\left(\frac{2}{p}\right)$ and $\left(\frac{q}{p}\right)$ next lecture.

Theorem 4.6: Let p be an odd prime. Then

$$\left(\frac{-1}{p}\right) = (-1)^{(p-1)/2} = \begin{cases} 1 & \text{if } p \equiv 1 \pmod{4} \\ -1 & \text{if } p \equiv 3 \pmod{4} \end{cases}.$$

Proof: The first equality follows from Euler's Criterion with $a = -1$. To prove the second equality set $p = 4k+1$ (resp $p = 4k+3$) for some $k \in \mathbb{Z}$, and compute $(-1)^{(p-1)/2}$.

□

Example: • $\left(\frac{-1}{7}\right) = -1$ by Theorem 4.6 since $7 \equiv 3 \pmod{4}$.

- $\left(\frac{-1}{53}\right) = 1$ by Theorem 4.6 since $53 \equiv 1 \pmod{4}$.

Lemma 4.7 (Gauss Lemma): Let p be an odd prime and let $a \in \mathbb{Z}$ with $p \nmid a$. Let n be the number of least positive residues of the integers $a, 2a, 3a, \dots, (\frac{p-1}{2})a$ that are greater than $p/2$. Then

$$\left(\frac{a}{p}\right) = (-1)^n.$$

Example: Use Gauss Lemma to find $\left(\frac{6}{11}\right)$.

We have $\left(\frac{6}{11}\right) = (-1)^n$ where n is the number of least positive residues of the integers:

$$6, 2 \times 6, 3 \times 6, 4 \times 6, 5 \times 6$$

that are greater than $11/2 = 5.5$. We have

$6 \equiv \boxed{6} \pmod{11}$	Thus
$2 \times 6 \equiv 12 \equiv 1 \pmod{11}$	$\left(\frac{6}{11}\right) = (-1)^3 = -1$
$3 \times 6 \equiv 18 \equiv \boxed{7} \pmod{11}$	by Gauss
$4 \times 6 \equiv 24 \equiv 2 \pmod{11}$	Lemma.
$5 \times 6 \equiv 30 \equiv \boxed{8} \pmod{11}$	